mathongo

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir

JEE Mains 2020 Jan Chapter wise Question Bank

Definite Integration

Q1

Given $f(a+b+1-x) = f(x)\ \forall\ x \in R$ then the value of $\frac{1}{(a+b)}\int_a^b x[f(x)+f(x+1)]dx$ is equal to

दिया गया है $f(a+b+1-x) = f(x)\ \forall\ x \in R$ तब $\frac{1}{(a+b)}\int_a^b x[f(x)+f(x+1)]dx$ का मान बराबर है–

(1) $\int_{a+1}^{b+1} f(x)dx$ (2) $\int_{a+1}^{b+1} f(x+1)dx$ (3) $\int_{a-1}^{b-1} f(x)dx$ (4*) $\int_{a-1}^{b-1} f(x+1)dx$

7th Jan Morning

Sol

(4)

$I = \frac{1}{(a+b)}\int_a^b x[f(x)+f(x+1)]dx$ ......(1)

$x \to a+b-x$

$I = \frac{1}{(a+b)}\int_a^b (a+b-x)[f(a+b-x)+f(a+b+1-x)]dx$

$I = \frac{1}{(a+b)}\int_a^b (a+b-x)[f(x+1)+f(x)]dx$ ....(2)

[$\because$ put $x \to x+1$ in given equation]

[$\because$दी गई समीकरण में $x \to x+1$ रखने पर]

(1) + (2)

$2I = \int_a^b [f(x+1)+f(x)]dx$

$2I = \int_a^b f(x+1)dx + \int_a^b f(x)dx$

$\int_a^b f(a+b+1-x)dx + \int_a^b f(x)dx$

$2I = 2\int_a^b f(x)dx$

02

Let $4\alpha\int_{-1}^{2} e^{-\alpha|x|}dx = 5$ then $\alpha =$

माना कि $4\alpha\int_{-1}^{2} e^{-\alpha|x|}dx = 5$ तो $\alpha =$

(1) $\ell n2$ (2) $\ell n\sqrt{2}$ (3) $\ell n\frac{3}{4}$ (4) $\ell n\frac{4}{3}$

7th Jan Evening

Sol

(1)

$$4\alpha\left\{\int_{-1}^{0} e^{\alpha x}dx + \int_{0}^{2} e^{-\alpha x}dx\right\} = 5$$

$$\Rightarrow 4\alpha\left\{\left(\frac{e^{\alpha x}}{\alpha}\right)_{-1}^{0} + \left.\frac{e^{-\alpha x}}{-\alpha}\right|_{0}^{2}\right\} = 5$$

$$\Rightarrow 4\alpha\left\{\left(\frac{1-e^{-\alpha}}{\alpha}\right) - \left(\frac{e^{-2\alpha}-1}{\alpha}\right)\right\} = 5 \qquad \Rightarrow 4(2 - e^{-\alpha} - e^{-2\alpha}) = 5 \quad \text{Put } e^{-\alpha} = t$$

$$\Rightarrow 4t^2 + 4t - 3 = 0 \qquad \Rightarrow (2t + 3)(2t - 1) = 0$$

$$\Rightarrow e^{-\alpha} = \frac{1}{2} \qquad \Rightarrow \alpha = \ln 2$$

03

Let माना $I = \int_{1}^{2} \frac{1dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$ then तब

(1) $\frac{1}{9} < I^2 < \frac{1}{8}$

(2) $\frac{1}{3} < I^2 < \frac{1}{2}$

(3) $\frac{1}{9} < I < \frac{1}{8}$

(4) $\frac{1}{3} < I < \frac{1}{2}$

8th Jan Evening

Sol

(1)

$$f(x) = \frac{1}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$$

$$f'(x) = \frac{-1}{2} \frac{(6x^2 - 18x + 12)}{(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$= \frac{-6(x-1)(x-2)}{2(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$f(1) = \frac{1}{3}, \qquad f(2) = \frac{1}{\sqrt{8}}$$

$$\frac{1}{3} < I < \frac{1}{\sqrt{8}}$$

04

Find the value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$

$\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$ का मान है–

(1) π^2 (2) $2\pi^2$ (3) $3\pi^2$ (4) $4\pi^2$

9^{th} Jan Morning

Sol

(1)

$$\int_0^{\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} + \frac{(2\pi - x)\sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= \int_0^{\pi} \frac{2\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= 2\pi \int_0^{\pi/2} \frac{\sin^8 x}{\sin^8 x + \cos^8 x} + \frac{\cos^8 x}{\sin^8 x + \cos^8 x} dx$$

$$= 2\pi \int_0^{\pi/2} 1\, dx = 2\pi \times \frac{\pi}{2} = \pi^2$$

05

If $f(x) = a + bx + cx^2$ where a, b, c $\in R$ then $\int_0^1 f(x)dx$ is

यदि $f(x) = a + bx + cx^2$ जहां a, b, c $\in R$ तब $\int_0^1 f(x)dx$ बराबर है

(1) $\frac{1}{3}\left(f(1)+f(0)+2f\left(\frac{1}{2}\right)\right)$

(2) $\frac{1}{6}\left(f(1)+f(0)+4f\left(\frac{1}{2}\right)\right)$

(3) $\frac{1}{6}\left(f(1)+f(0)-4f\left(\frac{1}{2}\right)\right)$

(4) $\frac{1}{6}\left(f(1)-f(0)-4f\left(\frac{1}{2}\right)\right)$

9th Jan Morning

Sol

(2)

$$\int_0^1 (a+bx+cx^2)dx = ax + \frac{bx^2}{2} + \frac{cx^3}{3}\Big|_0^1 = a + \frac{b}{2} + \frac{c}{3}$$

$f(1) = a + b + c$

$f(0) = a$

$f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{C}{4}$

Now $\frac{1}{6}\left(f(1)+f(0)+4f\left(\frac{1}{2}\right)\right)$

$= \frac{1}{6}\left(a+b+c+a+4\left(a+\frac{b}{2}+\frac{c}{4}\right)\right)$

$= \frac{1}{6}(6a+3b+2c) = a + \frac{b}{2} + \frac{c}{3}$